

KILLARA HIGH SCHOOL

YEAR 12 MATHEMATICS EXTENSION 1

TERM 1 ASSESSMENT

2004

Time Allowed: 2 hours

72 marks

Approved calculators may be used.

Show ALL necessary working.

Marks may not be awarded for untidy work.

Question 1 (10 marks)

Marks

(a) Give exact values for

(i) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$

1

(ii) $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

1

(b) If $\sin \alpha = \frac{3}{4}$, $0 < \alpha < \frac{\pi}{2}$ and $\sin \beta = \frac{2}{3}$, $\frac{\pi}{2} < \beta < \pi$, find the exact values of

(i) $\tan 2\alpha$

2

(ii) $\cos(\alpha - \beta)$

3

(c) Find the exact value of $\sin\left(2 \tan^{-1} \frac{1}{\sqrt{5}}\right)$

3

Question 2 (12 marks)**Marks**

(a) Simplify the following:

(i) $\frac{d}{dx}(\tan x)$ 1

(ii) $\frac{d}{dx}(\tan^2 x)$ 1

(iii) $\int \cos^2 x \, dx$ 2

(iv) $\int \sin x \cos^2 x \, dx$ 3

(v) $\int 5^x \, dx$ 2

(b) Using the substitution $u=1+t$, find $\int \frac{t}{\sqrt{1+t}} dt$ 3

Question 3 (12 marks)

(a) Using the substitution $u=2-x$, evaluate $\int_1^2 x\sqrt{2-x} \, dx$ 5

(b) r and θ vary in such a way that the area of the sector POQ has a constant value of 100cm^2 ; show that $\theta = \frac{200}{r^2}$ 2

(c) Using the result $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, find $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$ 3

(d) Simplify $\int \frac{6x+12}{x^2+4x} dx$ 2

Question 4 (12 marks)

(a) For the function $y=(x-1)^2-3$

(i) sketch the curve, showing all important features 3

(ii) find the largest possible domain such that this function will have an inverse 2

(b) (i) Find the domain and range of $y=\cos^{-1}2x$ 2

(ii) Sketch the curve in (i). 2

Question 4 continued:

Marks

(c) For the function $y = \sqrt{2-x}$

(i) find $\frac{dy}{dx}$ 1

(ii) make x the subject and hence find $\frac{dx}{dy}$. Hence... 1

(iii) show that $\frac{dy}{dx} \times \frac{dx}{dy} = 1$ 1

Question 5 (12 marks)

(a) (i) Evaluate $\frac{3\pi - \ln 3}{\sqrt{\pi} + e}$ to 3 significant figures. 2

(ii) Express as a single logarithm: $5 \log x + 3 \log y - \frac{1}{3} \log z$ 2

(b) Differentiate (i) e^{6x+3} 1

(ii) $e^{2x^3} + \log 3x$ 2

(iii) $5x - \log 2x^4$ 2

(iv) $\ln \frac{e^x + 1}{e^x - 1}$ 3

Question 6 (13 marks)

(a) (i) Solve $2 \cos^2 \theta + 3 \cos \theta - 2 = 0$ for $0 \leq \theta \leq 2\pi$ 2

(ii) What is the general solution of this equation in (i)? 1

(b) Write down the range of values for $y = e^{\sin x}$ and find the smallest possible positive value of x for which it is to be a minimum. 3

(c) Show that $\frac{x+7}{x-1}$ can be written as $1 + \frac{8}{x-1}$ and hence show that $\int_2^{e+1} \frac{x+7}{x-1} dx = e + 7$. 3

(d) The arc of the curve $y = \log_e x$ from $x=1$ to $x=e$ is rotated about the y axis. Find the volume of the resultant solid. 4

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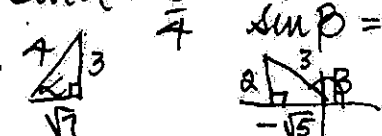
2004 Term 1 Worked Solutions

Year 12 Extension 1

1. (a) (i) $\sin^{-1} \frac{1}{\sqrt{2}} = \frac{\pi}{4}$ ✓ 45°?

(ii) $\tan^{-1}(-\frac{1}{\sqrt{3}}) = -\frac{\pi}{6}$ ✓ -30°?

(b) $\sin \alpha = \frac{3}{4}$ $\sin \beta = \frac{2}{3}$



(i) $\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$
 $= \frac{2 \times \frac{3}{4}}{1 - \frac{9}{16}}$

$= \frac{6}{\frac{7}{16}} \div -\frac{2}{7}$

$= -\frac{42}{2\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}}$

$= -\frac{42\sqrt{7}}{2 \times 7}$

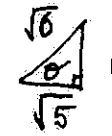
$= -3\sqrt{7}$ ✓

(ii) $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$ ✓

$= \frac{\sqrt{7}}{4} \times -\frac{5}{3} + \frac{3}{4} \times \frac{2}{3}$ ✓

$= -\frac{\sqrt{35}}{12} + \frac{1}{2}$ ✓

$= \frac{6 - \sqrt{35}}{12}$

(c) $\sin(2 \tan^{-1} \frac{1}{\sqrt{5}})$  ✓

$= \sin 2\theta$

$= 2 \sin \theta \cos \theta$ ✓

$= 2 \times \frac{1}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$ ✓

$= \frac{2\sqrt{5}}{\sqrt{5}}$
 $= \frac{2}{1} = 2$

2. (i) $\frac{d}{dx}(\tan x) = \sec^2 x$ ✓

(ii) $\frac{d}{dx}(\tan^2 x) = \sec^2 x \cdot 2 \tan x$
 $= 2 \tan x \sec^2 x$ ✓

(iii) $\int \cos^2 x dx = \frac{1}{2} \int (\cos 2x + 1) dx$ ✓
 $= \frac{1}{2} (\frac{1}{2} \sin 2x + x + c)$
 $= \frac{1}{4} \sin 2x + \frac{1}{2} x + c$ ✓

(iv) $y = \cos^3 x$
 $y' = -3 \cos^2 x \times \sin x$ ✓

$\therefore \int \sin x \cos^2 x dx$

$= -\frac{1}{3} \int -3 \sin x \cos^2 x dx$ ✓

$= -\frac{1}{3} \cos^3 x + c$ ✓

(v) $\int 5^x dx$ $5 = e^{\ln 5}$
 $= \frac{1}{\ln 5} \times 5^x + c$ ✓

$= \frac{5^x}{\ln 5} + c$ ✓

10

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$$2(b) \int \frac{t}{\sqrt{1+t}} dt \quad \begin{array}{l} u=1+t \\ \frac{du}{dt} = 1 \\ dt = du \\ t = u-1 \end{array}$$

$$= \int \frac{t}{u^{1/2}} dt$$

$$= \int \frac{u-1}{u^{1/2}} du$$

$$= \int (u^{1/2} - u^{-1/2}) du$$

$$= \frac{2}{3} u^{3/2} - 2u^{1/2} + C$$

$$= \frac{2}{3} (1+t)^{3/2} - 2(1+t)^{1/2} + C$$

$$3.(a) \int_1^2 x\sqrt{2-x} dx \quad \begin{array}{l} u=2-x \\ x=2-u \\ \frac{du}{dx} = -1 \\ du = -dx \\ u=2-2=0 \\ u=2-1=1 \end{array}$$

$$= \int_1^2 (2-u) u^{1/2} (-du)$$

$$= - \int_1^2 (2u^{1/2} - u^{3/2}) du$$

$$= - \int_0^1 (2u^{1/2} - u^{3/2}) du$$

$$= \left[\frac{2u^{3/2}}{3/2} - \frac{u^{5/2}}{5/2} \right]_0^1$$

$$= \left[\frac{4u^{3/2}}{3} - \frac{2u^{5/2}}{5} \right]_0^1$$

$$= \frac{4}{3} - \frac{2}{5} - (0)$$

$$= \frac{14}{15}$$

$$(b) \text{Area Sector} = \frac{1}{2} r^2 \theta_c$$

$$100 = \frac{1}{2} r^2 \theta_c$$

$$\frac{200}{r^2} = \theta$$

$$\text{i.e. } \theta = \frac{200}{r^2}$$

$$3(c) \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1 - (1 - 2\sin^2 x)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{2\sin^2 x}{x^2}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{\sin x}{x}$$

$$= 2 \times 1 \times 1$$

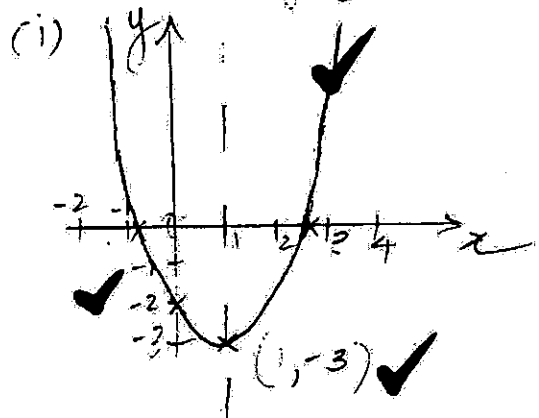
$$= 2$$

$$(d) \int \frac{6x+12}{x^2+4x} dx$$

$$= 3 \int \frac{2x+4}{x^2+4x} dx$$

$$= 3 \ln(x^2+4x) + C$$

$$4.(a) y = (x-1)^2 - 3 \quad \begin{array}{l} x=0 \\ y=-2 \\ \text{vertex } (1, -3) \\ \text{axis of symmetry } x=1 \end{array}$$

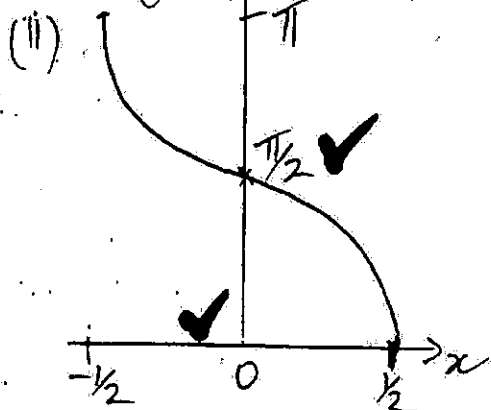


$$(ii) x \leq 1$$

$$\text{or } x \geq 1$$

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7 (b) $y = \cos^{-1} 2x$



domain $-\frac{1}{2} \leq x \leq \frac{1}{2}$ ✓

range $0 \leq y \leq \pi$ ✓

(c) $y = \sqrt{2-x} = (2-x)^{1/2}$

(i) $\frac{dy}{dx} = \frac{1}{2}(2-x)^{-1/2} \times -1$

$= -\frac{1}{2\sqrt{2-x}}$ ✓

(ii) $y^2 = 2-x$

$x = 2-y^2$

$\frac{dx}{dy} = -2y$ ✓

(iii) LHS $= \frac{dy}{dx} \times \frac{dx}{dy}$

$= \frac{-1}{2\sqrt{2-x}} \times -2y$

$= \frac{-1}{2\sqrt{2-x}} \times -2\sqrt{2-x}$

$= +1$ ✓

$= \text{RHS}$ ✓

5 (a)(i) $\frac{3^{\pi} - \ln 3}{\sqrt{\pi} + e}$

$= \frac{30.44}{4.44}$ ✓
✓

5 (a)(ii) $\leq \log x + 3 \log y = \frac{1}{3} \log z$

$= \log x + \log y^3 = \log z^{1/3}$ ✓

$= \log \frac{x^3 y^3}{\sqrt[3]{z}}$ ✓

(b)(i) $\frac{d}{dx} (e^{6x+3})$

$= 6e^{6x+3}$ ✓

(ii) $\frac{d}{dx} (e^{2x^3} + \log 3x)$

$= 6x^2 e^{2x^3} + \frac{3}{3x}$

$= 6x^2 e^{2x^3} + \frac{1}{x}$ ✓

(iii) $\frac{d}{dx} (5x - \log 2x^4)$

$= 5 - \frac{8x^3}{2x^4}$ ✓

$= 5 - \frac{4}{x}$ ✓

(iv) $\frac{d}{dx} \left(\ln \frac{e^x + 1}{e^x - 1} \right)$

$= \frac{d}{dx} (\ln(e^x + 1) - \ln(e^x - 1))$ ✓

$= \frac{e^x}{e^x + 1} - \frac{e^x}{e^x - 1}$ ✓

$= \frac{-2e^x}{e^{2x} - 1}$ ✓

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$$6(a)(i) \quad 2\cos^2\theta + 3\cos\theta - 2 = 0$$

$$2\cos^2\theta + 4\cos\theta - \cos\theta - 2 = 0$$

$$2\cos\theta(\cos\theta + 2) - (\cos\theta + 2) = 0$$

$$(2\cos\theta - 1)(\cos\theta + 2) = 0$$

$$\cos\theta = -2 \quad 2\cos\theta = 1$$

no solution $\cos\theta = \frac{1}{2}$

$$\therefore \theta = \frac{\pi}{3}, \frac{5\pi}{3}$$



$$(ii) \quad \theta = 2\pi n \pm \cos^{-1} \frac{1}{2} = 2\pi n \pm \frac{\pi}{3}$$

$$(b) \quad 1 \leq \sin x \leq 1$$

$$y = e^{\sin x}$$

$$y' = e^{\sin x} \cdot \cos x$$

minimum at $y' = 0$

$$\therefore 0 = \cos x \cdot e^{\sin x}$$

$$\cos x = 0, \quad x = \frac{\pi}{2}, -\frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$x = \frac{\pi}{2} \quad -\frac{\pi}{2} \quad \frac{\pi}{2}^+$$

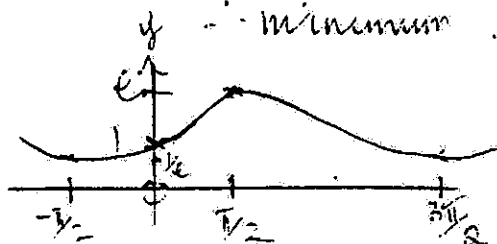
$$y' = ++ \quad 0 \quad +- \quad -$$

$\therefore$ maximum

$$x = -\frac{\pi}{2} \quad -\frac{\pi}{2} \quad -\frac{\pi}{2}^+$$

$$y' = +- \quad 0 \quad ++$$

$\therefore$ minimum



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$$x = \frac{3\pi}{2}$$

$$6(c) \quad 1 + \frac{8}{x-1}$$

$$t = \frac{x-1+8}{x-1}$$

$$= \frac{x+7}{x-1}$$

$$\therefore \int_2^{e+1} \frac{x+7}{x-1} dx$$

$$= \int_2^{e+1} \left(1 + \frac{8}{x-1}\right) dx$$

$$= \left[x + 8\ln(x-1) \right]_2^{e+1}$$

$$= e+1 + 8\ln(e+1-1) -$$

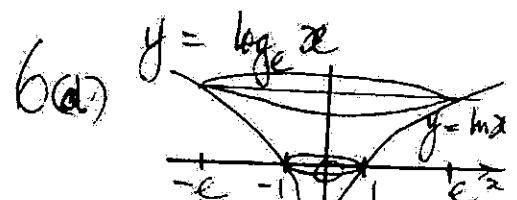
$$(2 + 8\ln(2-1))$$

$$= e+1 + 8\log_e e -$$

$$(2 + 8\log_e 1)$$

$$= e+1 + 8 - (2+0)$$

$$= e+7$$



$$V = \pi \int_0^1 x^2 dy$$

$$y = \log_e x \quad x = e, y = 1$$

$$e^y = x \quad x = 1, y = 0$$

$$V = \pi \int_0^1 e^{2y} dy$$

$$= \pi \left[\frac{1}{2} e^{2y} \right]_0^1$$

$$= \frac{\pi}{2} (e^2 - e^0)$$

$$= \frac{\pi}{2} (e^2 - 1)$$